GCAM SOFTWARE DOCUMENT SET

# GCAM-SW-SRS · Software requirements specification

> Demonstration document — generated from the repository's Markdown; not a regulatory submission.

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# VV.Studio.SRS — software requirements specification for GCAM Studio

Scope: what the three Studio projects (`Gcam.Studio.Core`, `Gcam.Studio.Services`, `Gcam.Studio`) must do. Not covered: the simulation engine and the original viewer `src/Gcam.Wpf` (removed in TODO-12 (2026-10-02); last present at `85b2ed1`). How the requirements are met is in VV.Studio.SDS (GCAM-SW-SDS); how they are verified, and their status, is in VV.Studio (GCAM-SW-VVR, 4. Verification).

Structured after IEC 62304 §5.2 (software requirements analysis). As in VV.Studio (GCAM-SW-VVR, 1. Scope and intended use), **GCAM Studio is not a medical device and no compliance is claimed**; the structure is borrowed for its discipline.

**At a glance**

- Active requirements (plus nine withdrawn rows) cover navigation, plotting, spectrum, acquisition, scene, image view, measurement, theme (functional, §3) and environment, accessibility, security, architecture (§4). Each is one testable present-tense statement.
- Inputs, outputs and every status message are listed with their valid ranges (§5); risk control (the illustrative safety class B) maps six hazardous situations to the requirements that control them (§6); what Studio deliberately does not do is in §7.
- Verification status per requirement is in VV.Studio §4 (GCAM-SW-VVR, 4. Verification).

## 1. Product context

| | |
|---|---|
| Purpose | A desktop viewer for Gcam, a Monte Carlo simulator of a coded-aperture gamma camera: place sources, run the simulation, inspect the detector flood map and the decoded reconstruction, measure on both in mm. Intended use and safety class: VV.Studio §1 (GCAM-SW-VVR, 1. Scope and intended use). |
| Users | Engineers and reviewers. They know what a flood map and a reconstruction are; they are not assumed to know the code. |
| Inputs | A scene of point sources (isotope, lateral X / Y, distance, activity), an absolute ambient field (dose rate and bound), preset live time, acquisition speed and an optional fixed Monte Carlo seed. Optics have editable physical defaults (`OpticsSettings`), frozen during acquisition and while data exist until Reset; decoder focus is an Imaging view setting. Inputs changed rarely (seed, detection chain, detector and environment inputs) are in the Options dialog (SR-RUN-12). |
| Outputs | Imaging flood/reconstruction with colour bars, found coordinates and measurements (distance, angle, ROI); Spectrum histogram/bands; Waveform traces/events; Detector face/gain geometry; retained-flood focus curves with a separate optional external range; acquisition status. Nothing is written to disk except the user-confirmed report export (SR-SEC-01, SR-EXPORT-01 … 05). |
| Upward trace | Studio is subsystem SS-4 of the product concept: it implements PR-SW-02 and serves user need UN-09 (engineering inspection; VV.Gcam.URS). |
| Neighbouring systems | The Gcam engine, reached through Core acquisition, spectrum, imaging, waveform, detector-face and focus-sweep service contracts (VV.Studio.SDS §3 (GCAM-SW-SDS, 3. Interfaces between items (§5.3.2, §5.4.3))); Windows (WPF, `WindowChrome` caption, UI Automation). |

## 2. Conventions

- **ID** `SR-<AREA>-<nn>`. IDs are stable: add, don't renumber; a withdrawn requirement keeps its row, marked *withdrawn*.
- Each row is one testable statement in the present tense ("is clamped", "keeps") — read it as *shall*.

- Requirements were derived from the code and the `DESIGN.*` pages as of 2026-10-01 (the code existed first; this is a reverse-engineered baseline, not a pre-development specification — §8).
- **Displayed text and help marks (TODO-44, 2026-10-12).** "Displayed", "shown", "visible text" and "labelled" include text behind a keyboard-focusable help mark ("?") at the right end of the row it explains, whose tooltip opens on hover and keyboard focus and whose automation HelpText carries the same text. Errors, refusals, experimental labels, state, progress, image / count roles and key numbers stay inline and never move behind a help mark.
- Requirements marked **RC** are risk-control measures (§6). Verification method, evidence and status per row are in the traceability matrix (GCAM-SW-VVR, Traceability matrix); allocation to software units is in VV.Studio.SDS §7 (GCAM-SW-SDS, 7. Requirement allocation).

# 3. Functional and capability requirements (§5.2.2 a)

### Workspaces (`SR-NAV`)

| ID | Requirement |
|---|---|
| SR-NAV-01 | One shared scene, optics, detector inputs, background ratio, live time, speed, seed, acquisition state and cumulative snapshot feed all registered workspaces; imaging owns its measurements and peak reading. |
| SR-NAV-02 | With at least two registered workspaces, a segmented switch exposes their titles and unique `Workspace.*` AutomationIds; activating a workspace through its checked state, click or Ctrl+1…5 changes the selected workspace. Unavailable indices do nothing, and a run retains selection. A single workspace has no switch. |
| SR-NAV-03 | Physical run inputs are editable only while no acquired data exist (SR-RUN-12); workspace view settings and selection are editable in every state and only re-project retained data. |

### Plot surface (`SR-PLOT`)

| ID | Requirement |
|---|---|
| SR-PLOT-01 | A series supplies finite Y with increasing X or origin + positive step, name, Line / Area / Histogram kind and colour role; bands and labelled X markers are available. Input is capped at 10 million samples per series. |
| SR-PLOT-02 | Linear and log10 Y map between pixels and data; log counts ≤1 clamp to 1. Linear ticks use 1–2–5 steps with the decimals the step needs and grouped thousands ("0 10 20", "1,000"), log decades have 2…9 minor ticks, and labels follow the engineering notation of the image colour scales. |
| SR-PLOT-03 | A pyramid built once per published dataset preserves exact per-column extrema, including short series, singleton data and boundary ranges; navigation does not rescan the whole trace. |
| SR-PLOT-04 | Plot X zoom / pan / reset have pointer and keyboard paths; read-only pointer readout, a focus ring, theme brushes and an automation peer are exposed. Resize retains X navigation. |
| SR-PLOT-05 | In a visible WPF window a 10-million-sample trace has CPU redraw ≤16 ms after a zoom and after a resize, including axes, query, geometry and drawing commands. Dispatcher / composition latency is recorded separately. |
| SR-PLOT-06 | Histogram data has N+1 finite increasing edges (or origin + positive width) and nonnegative counts. Bins at least two device pixels wide draw filled steps; narrower bins use exact column extrema. Zero bins occupy their full width at the floor. |
| SR-PLOT-07 | Same-full-range data updates preserve X zoom / pan. Changed ranges, cleared acquisitions and double-click / 0 / Home resets fit X. Log Y and resize preserve X. |

| ID | Requirement |
|---|---|
| SR-PLOT-08 | Y follows visible X with 8 percent linear headroom or the next half decade in log mode. Data updates only grow the top; navigation and log changes recompute it. |
| SR-PLOT-09 | Histogram hover highlights a half-open bin and draws its centre cursor; readout gives centre, bounds and counts with host units / formats. Default precision follows bin width. Line hover retains continuous coordinates. |
| SR-PLOT-10 | Band and marker labels sit in a label strip above the plot area, never over data: band labels centre on their bands, marker labels start right of their marker with a leader line, both clamp to the plot width and take additional collision rows. A plot may hide its marker labels (lines stay). A label wider than the plot is cut with an ellipsis on one line. |
| SR-PLOT-11 | A separate offscreen STA test, opt-in with GCAM_RENDER_SNAPSHOTS=1, renders fixed-spectrum dark / light, full / 662-keV zoom PNGs without windows, focus or mouse input. |

## Spectrum (`SR-SPEC`)

| ID | Requirement |
|---|---|
| SR-SPEC-01 | Spectrum displays a live stepped, filled histogram of the shared acquisition's measured event deposits, with measured energy in keV and acquired counts, on a fixed 0–2000 keV axis in 1000 bins of 2 keV for every scene, ambient field and pile-up setting. A pulse measured at or above 2000 keV is not drawn: it is counted as overflow, shown in the spectrum summary ("N overflow ≥ 2000 keV") and included in the total — Co-60 2505 keV cascade sums and pile-up sums above 2000 keV fall there. It adds no synthetic noise floor or independent event pool. |
| SR-SPEC-02 | Measured energy applies the acquisition's fixed pixel gain to its true deposit, then smears once through the default GAGG(Ce), S13360-3050 and CSP + CR-RC chain's `FrontEndModel`. The read-only chain resolution at 662 keV excludes pixel gain spread; no resolution override is offered. |
| SR-SPEC-03 | Optional pile-up sums gained amplitudes whose actual arrival gaps are below the chain-derived resolving time, extending the interval after each pulse, then smears the summed pulse once; toggling it reprocesses retained events. |
| SR-SPEC-04 | Each emission window spans E ± N·FWHM(E); adjacent lines separated by less than FWHM at their mean merge into one labelled band spanning their windows. Resolved lines keep separate bands even if their windows overlap. |
| SR-SPEC-05 | The emission table gives isotope, every grouped line energy, window limits, counts and share selected by bin centre; the total in-window share counts each bin once across all bands. A band whose lines are all X-rays in the engine isotope table is named by their emitter from that table with the source in brackets ("Ba K X-rays (Cs-137)"; the photons come from the daughter barium). Numbers are right-aligned under right-aligned headers that share measured numeric column widths and padding, with at least 8 DIP between adjacent cells at the supported window sizes. |
| SR-SPEC-06 | Log Y, positive finite window multiplier (default 1.5) and pile-up are view settings: they reuse acquired events without starting acquisition or changing the acquisition state. |
| SR-SPEC-07 | Same events, settings and seed produce identical histogram counts regardless of snapshot partition or pile-up toggle replay; new events are processed incrementally and CPU processing runs in a service worker. |
| SR-SPEC-08 | Spectrum exposes the plot and readout, table and right-panel controls with accessible names and AutomationIds `Spectrum.Plot`, `Spectrum.Lines`, `Spectrum.LogY`, `Spectrum.Window` and `Spectrum.PileUp`. |
| SR-SPEC-09 | Counts carry explicit bin edges; the graph uses keV and counts. Selecting an emission-window row requests its range plus one window width on each side, clamped to the axis. Snapshot replacement retains selection without another zoom request. |

## Acquisition (`SR-RUN`)

| ID | Requirement |
|---|---|
| SR-RUN-01 | *Withdrawn* — batch run / Succeeded; replaced by SR-RUN-09, SR-RUN-19. |
| SR-RUN-02 | *Withdrawn* — cancel / previous run retained; replaced by SR-RUN-10. |
| SR-RUN-03 | Any other exception becomes `State = Failed` and status "Failed: <message>"; it is never rethrown to the UI. |
| SR-RUN-04 | *Withdrawn* — photon progress; replaced by SR-RUN-11. |
| SR-RUN-05 | *Withdrawn* — batch input locking; replaced by SR-RUN-12. |
| SR-RUN-06 | *Withdrawn* — batch command availability; replaced by SR-RUN-13. |
| SR-RUN-07 | *Withdrawn* — batch result staleness; replaced by SR-RUN-14. |
| SR-RUN-08 | *Withdrawn* — photon-budget config; replaced by SR-RUN-15. |
| SR-RUN-09 | Start without acquired data executes fresh list-mode MC transport and decoding off the UI thread through `IAcquisitionService`, with the acquisition's Monte Carlo seed (SR-RUN-25); the state is Acquiring until Stopped, Completed or Failed. |
| SR-RUN-10 | Stop cooperatively ends the current acquisition segment, retains all acquired data, measurements and the session, and sets Stopped; physical inputs stay locked (SR-RUN-12). |
| SR-RUN-11 | Progress equals acquired live time / current preset live time, never decreases within a segment and reaches 1 at Completed; raising the preset (SR-RUN-24) lowers it accordingly. |
| SR-RUN-12 | Physical run inputs — sources (add / remove, isotope, X, Y, distance, activity), optics, readout mode, reflector gap, gain σ / seed, background ratio, ambient dose rate and bound, detection chain and the Monte Carlo seed — are editable only in Empty or after a failure without data, and not while a readout preparation runs. Any other change is refused by the view model, not only disabled in the view. While Acquiring, live time and speed are locked too. Sources, optics and the readout are edited in the left panel; the seed (Acquisition tab), the detection chain, the entrance / backing facts, reflector gap and gain σ / seed (Detector tab), photon H*(10), ambient bound and background ratio (Environment tab) in the **Options dialog**, an in-window modal layer opened from the top-bar cog. The dialog edits a session-only draft (never persisted): OK validates every field with the live inputs' rules and messages, and on any invalid field keeps the dialog open on that field's tab and writes nothing; Cancel or Escape discards the draft; no button is the default, so Enter in a field neither applies nor discards it. While the inputs are locked the dialog shows the values read-only with "Acquired settings · locked until Reset" and one Close button. Every Options value that differs from the Studio defaults (one Core record) is listed in the status bar's Options summary — with data followed by the acquired ambient BSR — and marked by a badge on the cog, so no hidden setting changes a result unseen. |
| SR-RUN-13 | Without acquired data, Start is enabled when at least one source exists or the ambient field is non-zero (a source-free, background-only acquisition; refused with the physical readout, SR-RUN-34). With data the button reads Continue and is enabled only in Stopped or Completed when the preset exceeds the acquired live time. The command re-checks its condition when invoked; it is not the window's default button. A Start refused by validation states its reason in the status bar in the warning colour — "Start refused: <reason>", or "Options: <reason>" for an Options input (seed, ambient dose or reflector gap; a field-only scene with the physical readout is a Start refusal, SR-RUN-34) — until the next started acquisition or Reset, keeps the field messages where they are, and leads to the input: it selects the left tab holding it (Optics for optics and reflector-gap errors, Scene for scene errors, Detector for readout refusals) or opens the Imaging focal-plane section for a decoder-focus error, and an Options refusal puts a warning state on the cog until the input is corrected; the dialog opens only on the user's click. |
| SR-RUN-14 | *Withdrawn* — outdated / stale result marking; replaced by SR-RUN-12, SR-RUN-23, SR-RUN-26. |

| ID | Requirement |
|---|---|
| SR-RUN-15 | The scene config uses nearest-prime rank, non-cyclic decoding and a reconstruction grid inside the FCFOV; non-finite or non-positive preset live time and speed are rejected by the service. |
| SR-RUN-16 | Immutable cumulative snapshots carry live time, integer counts, flood, reconstruction, estimate and the same fresh event list (pixel, true deposit in keV, Poisson arrival time in s); each event is used once. The flood adds one count at each ComptonCrystalDetector event's Argmax pixel, so in-crystal Compton scatter mispositioning is part of the image. A decay of an isotope with correlated gammas (Co-60 with its 1173 / 1332 keV angular correlation, Na-22 with its back-to-back annihilation pair) is one event: the deposits of all its detected gammas summed, at the largest-deposit pixel over their merged interaction sites, with one arrival time (true-coincidence summing). |
| SR-RUN-17 | At 4 Hz the accumulated flood is re-decoded on its fixed grid and imaging measurements refresh without replacing their geometry. A slow consumer receives the latest cumulative snapshot. |
| SR-RUN-18 | If MC cannot supply rate × speed, live time advances only through the acquired prefix and the status appends "MC-limited ×k" with achieved live seconds per wall second. |
| SR-RUN-19 | Preset live time defaults to 60 s and speed to ×10; reaching the preset automatically sets Completed. |
| SR-RUN-20 | Studio explicitly supplies entrance 0.15 mm steel-equivalent, backing 2 mm, default reflector gap 0.1 mm, gain σ 3% and gain seed 1, from one Core defaults record that also feeds the initial values, the invalid-input fallbacks and the Options summary (SR-RUN-12). Entrance and backing are read-only facts and gap, gain σ and gain seed editable run inputs on the Options dialog's Detector tab; gain σ / seed are disabled under the experimental physical readout, which does not use them. Existing engine scene-builder defaults are unchanged. |
| SR-RUN-21 | Snapshots retain their acquisition detector inputs. True deposits are preserved; one shared deterministic measurement response applies the CrystalUniformity gain pattern before chain smearing. Gain inputs are locked while data exist, so the recorded detector response cannot change. |
| SR-RUN-22 | A finite nonnegative detected background/source ratio (default 0) adds an independent Poisson process at BSR × source rate. Each fresh background deposit comes from the existing unmasked cosine-flux crystal response at 200 keV and is placed according to the uniform detected pedestal. At zero BSR the seeded source stream is unchanged. BSR is a pre-trigger detected-input ratio to the running source detected-rate estimate. With the physical readout each BSR input is a fresh unmasked front-entry cosine-flux history transported through the module (gaps, entrance slab, outgoing backing proxy) that keeps its own transported pixel and interaction sites — not the uniform pedestal; a history that scores nothing is retried at the scheduled time with bounded work; zero source rate gives zero BSR inputs (TODO-42 stage B, 2026-10-13). |
| SR-RUN-23 | Continue (Start with data) resumes the same acquisition: the same list-mode source and random streams, the already-drawn look-ahead event, the events, flood, live-time clock, seed and acquisition identity. Live time does not advance while stopped. The continued event stream equals, event for event, that of an uninterrupted acquisition to the same live time, and the workspaces append to their retained processing. |
| SR-RUN-24 | In Stopped and Completed the preset live time may be raised to continue; a value below the acquired live time is rejected with a message and the previous value restored. Speed only paces the acquisition and is editable in every state but Acquiring. |
| SR-RUN-25 | Each new acquisition draws a new Monte Carlo seed unless the Options seed fixes one (blank: new; a nonnegative integer: fixed). Every random stream of the acquisition derives from it, the same seed reproduces the acquisition, and the status line shows it. |

| ID | Requirement |
|---|---|
| SR-RUN-26 | Reset — enabled only in Stopped, Completed or Failed with data, never while acquiring — first asks for one deliberate confirmation in an in-window layer that names the acquired live time and counts and says the inputs unlock for the next acquisition; Cancel is the default button and Escape cancels, keeping the acquisition. Discard removes the session, snapshot, images, spectrum, scope, focus sweep and the scene frozen at Start; the state returns to Empty ("Ready") and the physical inputs unlock; measurement shapes are kept. *Reversed 2026-10-12 (TODO-44): Reset formerly asked no confirmation; the author: a discard needs one deliberate confirmation, Cancel being the default.* |
| SR-RUN-27 | A failure after data keeps the last published data visible and locked and offers only Reset; a failure before any data leaves the inputs editable and Start enabled. Both show "Failed: <message>". |
| SR-RUN-28 | The status line's rate is the observed count rate, counts / live time — the definition the Detector workspace uses. |
| SR-RUN-29 | An absolute ambient field — photon H*(10) in µSv/h, entered as digits with at most one decimal point (`^\s*(\d+(\.\d*)?\|\.\d+)\s*$`), default 0.10, with a bound, front-only through the mask (default) or bare crystal on all faces — adds the engine's source-independent ambient photon process (with the physical readout, transported with recorded interaction sites through the same module construction as the signal; SR-DET-05) using the validated terrestrial spectrum `terrestrial-unscear2000-v1`, the only offered preset. The dose rate and bound are entered on the Options dialog's Environment tab. Any other dose entry is refused with the message under the field (OK keeps the dialog open and applies nothing); the previous dose rate stays, and Start is refused with the reason (SR-RUN-13) until it is corrected. The spectrum ships with Studio and is used only if its bytes match the SHA-256 pinned in Studio and the engine accepts it as validated with its acceptance record; otherwise Start fails (SR-RUN-27). The Environment tab names the preset and the bound; 0 is labelled "ideal environment" and runs the unchanged ideal acquisition, bit for bit. A derived ambient BSR (ambient ÷ source detected rate) is shown read-only on the Environment tab and, with data, in the status bar's Options summary. |
| SR-RUN-30 | Physical response, overlap, trigger, hold, digitization, LUT assignment and calibrated energy shall be applied once before workspace processing. Immutable snapshots shall carry the measured prefix and calibration identity; workspace operations shall not reacquire or remeasure it. |
| SR-RUN-31 | Stop and Completed shall preserve readout RNG progression, transport look-ahead, pulse grid, active/queued inputs and trigger/hold/busy/re-arm state. A conversion shall be published only after its full hold search is observed. Continue shall be record-identical to uninterrupted acquisition at the same seed and final horizon, including an unfinished hold. |
| SR-RUN-32 | Readout transport/preparation shall run off the UI thread with cancellation. Acquisition snapshots shall use the existing nominal 4 Hz cadence and segment-boundary publications, show the consumed prefix and achieved speed under compute limits, and never invent future counts. View changes shall not repeat preparation or measurement. |
| SR-RUN-33 | Physical status counts/rate and Imaging All shall count valid LUT-assigned conversions; unknowns, all conversions and realised pre-trigger input hits shall be separately labelled. Spectrum shall histogram assigned conversions' stored calibrated energies with out-of-axis accounting. Explicit keV windows shall select the same inclusive record population in Spectrum and Imaging. Legacy gain/smearing, energy-only pile-up and stripping shall not be applied. The readout panel's Energy window section offers presets ±10, ±15, ±20 and ±25 % around the calibration line inside the current window (else 661.7 keV), with the benchmark's exact decimal rounding; a preset only writes the low / high fields, which stay editable, and the preset matching the current values is marked. Only ±15 % (the record's benchmark width) is labelled the EV-36 benchmark window; the presets' help states that the other widths have no measured localisation cost. |

| ID | Requirement |
|---|---|
| SR-RUN-34 | Physical Start, enforced in both shell and service before transport, shall require a matching successful preparation for every preparable calibration line the scene emits (exact energy equality) and shall refuse an unprepared, failed or stale line, a scene without a preparable line, and a field-only scene (ambient without a source — no field calibration line is offered; a product decision, not a physical limit) before transport, preserving entered values and displaying an actionable reason. A source scene with the ambient field (both bounds) and / or BSR runs the physical readout (TODO-42, 2026-10-13). Refused and incidental emissions remain in transport and do not block. Imaging-cost values and crystal diagnostics shall never gate an otherwise calibrated experimental readout. DirectCrystal inputs, seeded stream, defaults and microcell policy shall remain unchanged. |

### Scene (`SR-SCENE`)

| ID | Requirement |
|---|---|
| SR-SCENE-01 | Source inputs are clamped: distance 200–3000 mm, activity ≤ 0 → 1 µCi, unknown isotope → Cs-137; the list label follows every edit. |
| SR-SCENE-02 | Startup has one selected source. Adding selects the new source, offset so sources don't stack; removing selects the neighbour. |

### Image view (`SR-VIEW`)

| ID | Requirement |
|---|---|
| SR-VIEW-01 | Fit shows the whole image, aspect preserved, centred. |
| SR-VIEW-02 | At fit, each cell spans a whole number of **device** pixels (any display scaling), and the image stays centred. |
| SR-VIEW-03 | Zoom (wheel, + / −) keeps the point under the cursor fixed, is clamped to 1–32×, and full zoom-out returns to the centred fit. |
| SR-VIEW-04 | Pan (drag, arrows) cannot move the image out of view; at fit the image stays centred. |
| SR-VIEW-05 | Screen ↔ image ↔ mm mapping: row 0 at the bottom (y up), pixel *i* centred at `origin + i·step` mm, round trip exact. |
| SR-VIEW-06 | Resizing keeps the zoom; a new image size refits. |
| SR-VIEW-07 | The hovered pixel is read out as "x … mm, y … mm · value unit", the value to four significant digits with grouped thousands (integers in full); double-click, `0` or Home refits. |
| SR-VIEW-08 | After a run, the reconstruction header shows the decoded peak in mm; when the selected channel has two or more found peaks it shows their number ("2 peaks found") instead of one of them. |
| SR-VIEW-09 | The colour bar and readout under an image span the width the image is drawn at (after whole-device-pixel cell snapping), as do the Detector face's colour bar and legend. |

### Per-nuclide imaging (`SR-IMG`)

| ID | Requirement |
|---|---|
| SR-IMG-01 | One shell-owned positive finite window multiplier N (default 1.5) is editable in Spectrum and Imaging. Both use the same per-energy FWHM bands, including unresolved-line grouping. Changing N re-filters retained acquisition events without a new acquisition. |
| SR-IMG-02 | The worker accumulates a primary-line energy-window flood per scene isotope, using the acquisition's deterministic measured energy response, and decodes it. All retains the acquisition image of every event. |

| ID | Requirement |
|---|---|
| SR-IMG-03 | An All / isotope selector changes the displayed flood, reconstruction, colour bar, readout, peak reading and ROI statistics together. Selecting a channel does not run transport or decoding on the UI thread. |
| SR-IMG-04 | Each isotope channel reports up to its source count of separated found peaks, refined with the pipeline-default sub-cell interpolation (applied to MLEM's λ as well, SR-IMG-07). Found peaks are read-only diamonds labelled "Found <isotope>" and listed as coordinates; true-source rings are display-only. Full chips fit within the visible image/control intersection, with at least 4 DIP between chips and clearance from marker crosshairs; displaced labels use neutral leaders and never move marker centres. A chip that cannot fit without overlap is omitted, retaining its coordinate/measurement list entry. Layout recalculates on resize/zoom/pan. All shows the union of found peaks. Association is checked below one reconstruction-grid diagonal; localization precision is measured separately and is not implied by the marker. |
| SR-IMG-05 | A Compton strip view toggle subtracts the simultaneous raw higher-channel floods per pixel; the **displayed** flood is clamped at zero (`max(0, low − Σ R·high)`), cross-correlation **decodes the signed difference**, and both methods report the signed net count Σ low − R·Σ high as `N ± σ net counts` (1σ, counting + calibration, including single-contaminant window overlap; "uncertainty unavailable" where the model does not support it, e.g. several contaminants); a negative net is shown as is, no detection verdict is implied, and the scalar model removes no ambient background. H-only calibration at acquisition start and after N changes uses at least 20,000 accepted events per contaminating isotope, with R = low-window / own-window counts. The panel lists each R and states the one-pass scalar model limitation: exact for a clean pair, approximate for 3+ overlapping contaminants. |
| SR-IMG-06 | Channel building, H-only calibration and decoding run on a worker and publish separate processing costs. The scene and optics frozen at Start determine processing for the whole acquisition, including continued segments. Cancelled or late processing cannot replace a newer view. |
| SR-IMG-07 | A Reconstruction selector (Cross-correlation, the default; MLEM with a pixel-area forward model) re-decodes the retained All and isotope floods without transport, measurement, random draws or calibration; the latest revision wins. MLEM uses each channel's primary-line closed-cell transmission (All: the highest primary line) and a fixed 400 iterations, chosen by the pair-resolution rule at the default optics and confirmed on a third, disjoint seed set. With Compton strip, MLEM decodes the raw window flood with the higher lines' downscatter as a known background and reports the net count Σ low − R·Σ high, labelled as the net count with its uncertainty for both methods (SR-IMG-05); count and image-role labels follow the published result while a new method or strip setting is pending. The reconstruction readout unit is "(MLEM λ)". A note states the low-count false second peak (≈ 50 % of 250-count frames at quarter height), whether the iteration count was measured for the current optics and focal plane, and that no ambient background term is used. Measurements and the focus sweep are kept across a switch; the focus sweep always cross-correlates. With MLEM selected the display refresh drops to about 1 Hz at the default optics with five channels; transport is unaffected. |

## Measurement (`SR-MEAS`)

| ID | Requirement |
|---|---|
| SR-MEAS-01 | Distance is the Euclidean length in mm, shown with 0.1 mm resolution. |
| SR-MEAS-02 | Angle is measured at the vertex (second point), 0–180°; a zero-length arm gives 0. |
| SR-MEAS-03 | ROI counts whole pixels **by centre**, corners in any order, clipped to the image; sum, mean and max. The value carries its pane's unit: on a stripped flood it is a statistic of the displayed (clipped) values, on a reconstruction a decoded / MLEM λ statistic; neither replaces the channel-wide net count. |
| SR-MEAS-04 | ROI values are recomputed on every new result; with no data on that pane the value is "—". |

| ID | Requirement |
|---|---|
| SR-MEAS-05 | Measurements are numbered M1, M2, …; a new one is selected; deleting one — its row's delete button (shown on the selected row and under the pointer) or the Delete key — removes that row; removing the selected row selects its neighbour, removing another keeps the selection; the header's clear-all (trash can) restarts numbering; a draft with the wrong point count is rejected. |
| SR-MEAS-06 | The pointer tools — Pan (default), Distance, Angle, ROI — are icon toggles in the Measurements header, exactly one active; the tool hint follows the active tool. |
| SR-MEAS-07 | Gestures: drags shorter than 4 px are ignored; Esc or right-click abandons a draft; Delete on the focused heatmap removes the selected measurement; points are clamped to the image. Geometry is stored in mm, so it follows zoom and pan. |
| SR-MEAS-08 | *Withdrawn* — source-marker drag; source positions are physical inputs (SR-RUN-12), edited in the left panel. |

### Theme (`SR-THEME`)

| ID | Requirement |
|---|---|
| SR-THEME-01 | The top-bar theme icon button (sun / moon) switches dark ↔ light; its tooltip and accessible name name the theme it switches *to*. |
| SR-THEME-02 | Switching replaces the token dictionary in place and every window repaints (`DynamicResource`), the caption included. The window has no OS title bar (`WindowStyle="None"` with `WindowChrome`): the top bar is the caption — drag to move, double-click to maximise / restore, OS snap, resize borders, Alt+Space and Alt+F4 — and themed caption buttons Minimize, Maximize / Restore and Close sit flush at the top right at the bar height, frameless at rest, Close hovering red, each with a tooltip, an accessible name and keyboard focus. A maximised window fits the monitor work area. DWM is asked to match any remaining native frame, best effort. The Windows 11 snap-layout flyout on the maximise button is not provided. *Revised 2026-10-13 (TODO-45): the OS title bar, which followed the theme through DWM, was removed; the author: it duplicated the top bar and ignored the theme.* |

## Editable optics and decoder focus

| ID | Requirement |
|---|---|
| SR-OPT-01 | Rank, cell pitch, mask–detector distance, pixels per side and pixel pitch are editable physical run inputs on the left panel's Optics tab. One validated effective record feeds configuration and preset matching. The prime selector displays the effective supported rank. Physical editors are editable only while no acquired data exist (SR-RUN-12). |
| SR-OPT-02 | Sharp (default), Baseline, Wide FOV and High-res apply their historical engineering geometries atomically. The numeric fields are always shown: a preset writes its values into them and disables them; Custom, chosen only through the selector, enables them starting from the current values, and edits in Custom never re-select a preset. Selecting a geometry does not move scene sources or alter decoder focus; performance is conditional on scene, focus, counts and isotope channels. *Revised 2026-10-13 (TODO-45): an edit under a preset formerly switched the selector to Custom or a matching preset, and the first stage-2 build showed the fields only for Custom; the author reversed the latter: fields that appear and vanish with a mode hide what the preset means.* |
| SR-OPT-03 | Studio rejects non-finite, malformed, out-of-policy and cross-field-invalid inputs before transport or projection. Policy ranges are in §5; every scene source lies beyond the mask front face. Refocus uses the acquired physical settings. Grid allocation is bounded at 128×128. Errors stay visible under their fields and at the foot of the left panel whichever tab is selected; a refused Start states the error in the status bar and selects the tab holding the input (SR-RUN-13). |

| ID | Requirement |
|---|---|
| SR-OPT-04 | Decoder focal plane is an Imaging view setting. At Acquiring, Stopped and Completed it reprojects All and isotope/stripped reconstructions, estimates and found peaks from retained floods/events without new transport, measurement, random draws or calibration. Latest revision wins; the acquisition state is unchanged. Focus changes clear reconstruction measurements/drafts with a visible explanation while retaining flood measurements. All found markers remain the union of isotope-channel markers. |
| SR-OPT-05 | Display resolution element cF/D, nominal cyclic field ±pcF/(2D), samples per mask cell, detector coverage in periods and physical mask width. These are geometry values, not localization/usable-field guarantees; no pass/fail glyphs or depth-reach extrapolation. A one-line caption states that precision is position-dependent conditional evidence; the 1 m position-sweep figures are its tooltip and accessible help text, and are recorded in VV.Studio.Imaging (GCAM-SW-VVR). |
| SR-OPT-06 | The shared left panel is a tab control — Scene (sources, selected source), Optics (engineering geometry: preset and fields) and Detector (the readout panel's sections) — and each tab's default content fits 1280×800 without scrolling; the detection chain and the detector and environment inputs are in the Options dialog (SR-RUN-12). The Imaging right panel holds collapsible Decoder focal plane, Imaging channel and Focus sweep sections (the sweep opens when a result arrives) and scrolls when needed; the Measurements panel (tool toggles in its header, table, selected-row detail) sits under the image pair and takes the height the width-limited images leave. Every vertically scrolling panel reserves a scrollbar gutter, so the bar keeps a gap from the content and the panel border. Controls inherit theme tokens and have accessible labels, units and stable AutomationIds. *Revised 2026-10-13 (TODO-45): the collapsible Detection chain / Physical optics / Detector sections with one-line summaries were replaced by tabs; the author: a long stack of expanders hid inputs and read as one block.* |

# 4. Non-functional requirements

## Operating environment (`SR-ENV`, §5.2.2 a, j)

| ID | Requirement |
|---|---|
| SR-ENV-01 | Studio runs on Windows 10 / 11 with the .NET 9 desktop runtime; the DWM dark-frame attribute for the remaining native frame is best effort (a Windows that lacks it ignores it without error). |
| SR-ENV-02 | The layout is designed for a 1440 × 900 work area; on a smaller work area the window starts maximised. |

## Usability and accessibility (`SR-A11Y`, §5.2.2 g)

| ID | Requirement |
|---|---|
| SR-A11Y-01 | Every interactive element has `AutomationProperties.Name`; each measurement row is named by its description ("M2 ROI on Flood: Σ 1,234"). |
| SR-A11Y-02 | `HeatmapView` has an automation peer: control type Image, class `HeatmapView`, keyboard-focusable, `ItemStatus` = zoom + readout. |
| SR-A11Y-03 | Keyboard paths: heatmap zoom / pan / fit by key; measurement tool access keys Alt+P / D / A / R (Pan / Distance / Angle / ROI) whenever the Imaging workspace is shown and no modal layer is open, named in each tool's tooltip and exposed as its accessible access key; sources movable via X / Y fields. Enter starts nothing: Start is not the default button (SR-RUN-13), and the Options dialog has no default button (SR-RUN-12). |
| SR-A11Y-04 | The status line is a polite live region; state and selection are never signalled by colour alone. |
| SR-A11Y-05 | Opaque retained-value disabled text has contrast ≥4.5:1 against Surface, Canvas and Raised in both themes, while remaining dimmer than Secondary/Primary text on Surface. Locked inputs retain their disabled state and subdued fill/borders. Disabled action templates with separate opacity are outside this retained-value target. |

### Security and data (SR-SEC, §5.2.2 e, h)

| ID | Requirement |
|---|---|
| SR-SEC-01 | Studio opens no network connection and reads or writes no user files except the report export (SR-EXPORT-01 … 05), which writes only the file at the path the user confirms and its declared sidecars (two files; three for event data). Scene, results and measurements otherwise live in memory only and are gone when the app closes. Studio keeps no settings. *Revised 2026-10-15 (TODO-46, D-58): the report export is the one declared write surface; stage 2 the same day added PDF, Word and event data.* |

### Architecture constraints (SR-ARCH)

Design constraints that make the other requirements verifiable (testability without a UI stack, one engine entry point). They are kept as requirements so that a change breaking them fails verification, not just review.

| ID | Requirement |
|---|---|
| SR-ARCH-01 | `Gcam.Studio.Core` references no WPF type. |
| SR-ARCH-02 | Only `Gcam.Studio.Services` references the simulation engine (`Gcam.Simulation`); the UI reaches acquisition, spectrum, imaging, waveform, detector-pattern and focus processing through their Core service contracts. |
| SR-ARCH-03 | View code-behind is `InitializeComponent()` only; views use theme keys, not literal colours or sizes. |
| SR-ARCH-04 | Studio tests reference `Gcam.Studio.Core` only (no WPF in tests). |

Seven withdrawn rows are retained with stable IDs; Detector, focus and Waveform requirements extend this baseline below.

## 5. Inputs, outputs, messages (§5.2.2 b–d)

### Inputs and their valid ranges

| Input | Unit | Valid range / handling | Requirement |
|---|---|---|---|
| Isotope | — | one of the engine's `Isotopes.All`; anything else → Cs-137 | SR-SCENE-01 |
| Source X, Y | mm | free in the fields while editable | SR-SCENE-01, SR-RUN-12 |
| Source distance | mm | 200–3000 (clamped) | SR-SCENE-01 |
| Activity | µCi | > 0 (≤ 0 → 1) | SR-SCENE-01 |
| Preset live time | s | finite > 0; default 60; invalid UI input → 60, invalid service input rejected; with data only ≥ the acquired live time | SR-RUN-15, SR-RUN-19, SR-RUN-24 |
| Speed | live s / wall s | finite > 0; default 10; invalid UI input → 10, invalid service input rejected | SR-RUN-15, SR-RUN-19, SR-RUN-24 |
| Seed (Options › Acquisition) | integer | blank (default: a new seed per acquisition) or a nonnegative integer; OK refuses anything else with the field message; an invalid value refuses Start ("Options: …") | SR-RUN-25, SR-RUN-12 |

| Input | Unit | Valid range / handling | Requirement |
|---|---|---|---|
| Gain σ / seed (Options › Detector) | % / integer | σ finite ≥ 0, default 3%; seed an integer, default 1; OK refuses invalid text with the field message; a non-finite or negative σ set programmatically → 3%; disabled under the physical readout | SR-RUN-20, -21 |
| Reflector gap (Options › Detector) | µm editor, mm configuration | finite 0 ≤ gap < pending pixel pitch; default 100 µm; OK refuses invalid text; revalidated after pitch edits, and a gap that no longer fits refuses Start ("Options: …", Optics tab selected) | SR-DET-01 |
| Focus sweep K | integer | 1–4, default 1; 81 planes uniform in inverse detector-referenced distance from D+30 to 3000 mm | SR-FOCUS-01, -02 |
| External surface range | mm from detector | optional, finite positive measurement; Use as focus requires valid decoder focus | SR-FOCUS-05 |
| Background BSR (Options › Environment) | detected background / source | finite ≥ 0, default 0; OK refuses invalid text; a non-finite or negative value set programmatically → 0; invalid service input rejected; available in both readout modes (the physical readout transports BSR histories through the module, SR-RUN-22) | SR-RUN-22, SR-RUN-34 |
| Ambient photon H*(10) (Options › Environment) | µSv/h | digits with at most one decimal point (no sign, exponent, NaN, ∞); default 0.10; 0 = ideal environment; an invalid entry is refused with the field message, the previous value stays and Start is refused | SR-RUN-29 |
| Ambient bound (Options › Environment) | — | front-only through the mask (default) or bare crystal on all faces (DirectCrystal); with the physical readout the second bound is the unmasked all-face module approximation (module gaps, entrance slab on front entry, backing slab on rear entry, no side housing) — not a bare-crystal upper bound; locked while data exist | SR-RUN-29 |
| Physical optics (Optics tab) | mm / integer | ranks 5, 7, 11, 13, 17, 19, 23; N integer 4–64; finite D ≥1, pitches ≥0.05; pixel pitch > reflector gap; source z > D+5 (10 mm slab); always shown, editable only for Custom (a preset fills and disables them); invalid edits refuse Start | SR-OPT-01, -02, -03 |
| Energy window (physical readout) | keV, inclusive | finite 0 ≤ low < high, default 600–720; presets ±10 / ±15 (EV-36 benchmark) / ±20 / ±25 % around the window line write both fields; an invalid pair is refused and the previous window stays | SR-RUN-33 |
| Decoder focus | mm from detector | finite, > acquired D; projected grid ≤128 cells per side; invalid edits retain the last valid view | SR-OPT-03, -04 |
| Spectrum window N | × FWHM(E) | positive finite; default 1.5; invalid numeric value → 1.5 | SR-SPEC-04, -06 |
| Spectrum log Y / pile-up | boolean | defaults true / false; view settings | SR-SPEC-03, -06 |
| Measurement points | mm | clamped to the image extent; 2 points (distance, ROI) or 3 (angle) | SR-MEAS-05, SR-MEAS-07 |

## Outputs

| Output | Format | Requirement |
|---|---|---|
| Flood map, reconstruction | heatmap (viridis, linear min–max) + colour bar; row 0 at the bottom | SR-VIEW-01…06 |

| Output | Format | Requirement |
|---|---|---|
| Hovered pixel | "x 1.2 mm, y −3.4 mm · 56.7" | SR-VIEW-07 |
| Decoded peak | "peak (x, y) mm", 0.1 mm | SR-VIEW-08 |
| Spectrum | stepped histogram on a fixed 0–2000 keV axis (2 keV bins) with an overflow count, measured energy (keV) / acquired counts; line bands and emission-window table | SR-SPEC-01, -04, -05 |
| Front-end chain | read-only chain name, resolution FWHM % at 662 keV and resolving time in ns | SR-SPEC-02, -03 |
| Distance / angle / ROI | "12.3 mm" · "67.0°" · "Σ 1,234" with px count, mean, max | SR-MEAS-01…04 |

## Status messages and warnings

| Situation | What the user sees | Requirement |
|---|---|---|
| Ready (Empty) | "Ready" after launch and after Reset; Start | SR-RUN-26 |
| Acquiring | "t = 12.0 s of 60 s · 1,834 counts · 153 cps · seed 81236", live-time progress, Stop button; optional "MC-limited ×k"; the rate is counts / live time | SR-RUN-11, SR-RUN-18, SR-RUN-25, SR-RUN-28 |
| Completed | "Completed · t = 60.0 s of 60 s · <counts> counts · <rate> cps · seed <n>"; Continue enabled once the preset is raised; Reset | SR-RUN-19, SR-RUN-24 |
| Stopped | "Stopped · t = <t> s of 60 s · <counts> counts · <rate> cps · seed <n>"; acquired data retained; Continue, Reset | SR-RUN-10, SR-RUN-23 |
| Failed | "Failed: <message>"; with data only Reset | SR-RUN-03, SR-RUN-27 |
| Preset below the acquired live time | "The preset cannot be below the acquired live time (<t> s)." in the status bar; previous value restored | SR-RUN-24 |
| Invalid seed | "The seed must be blank (new each acquisition) or a nonnegative integer." under the field in the Options dialog (OK refused); an invalid seed held by the view model also shows it in the status bar | SR-RUN-25 |
| Invalid ambient dose rate | "The ambient dose rate must be a number ≥ 0 in µSv/h (digits and one decimal point, e.g. 0.1)." under the field in the Options dialog (OK refused); the previous value stays | SR-RUN-29 |
| Invalid Options draft | the field message under each invalid field; the dialog stays open on the first invalid field's tab and nothing is applied | SR-RUN-12 |
| Start refused by validation | "Start refused: <reason>" in the warning colour in the status bar until the next started acquisition or Reset; the left tab holding the input is selected (or the Imaging focal-plane section opens) | SR-RUN-13, SR-OPT-03 |
| Start refused for an Options input | "Options: <reason>" in the warning colour in the status bar; warning state on the top-bar cog until the input is corrected | SR-RUN-13 |
| Non-default options | "Options: seed 42 · gap 150 µm · …" at the right of the status bar (semibold), "· ambient BSR …" appended with data; a badge on the cog; with every option at its default, nothing before an acquisition and "ambient BSR …" with data | SR-RUN-12, SR-RUN-29 |

| Situation | What the user sees | Requirement |
|---|---|---|
| Readout preparation | "Preparing <line, stage and history progress>" with the preparation progress bar in the status bar; then "Readout ready: <lines> keV", "Readout preparation cancelled", or "Readout preparation failed: <reason>" / "Readout preparation refused: <reason>" in the warning colour, until the next preparation, Start or Reset | SR-DET-07 |
| Spectrum processing failure | "Spectrum failed: <message>"; acquired data retained | SR-SPEC-01, -07 |
| ROI before any run / outside the pixels | "—" · "no image yet" / "no pixel centres inside" | SR-MEAS-04 |

Every status is carried by text, not colour alone (SR-A11Y-04).

## 6. Risk-control requirements (§5.2.3)

Hazardous situations and their analysis are in VV.Studio §6 (GCAM-SW-VVR, Risk table (illustrative, IEC 62304 §7 style)). The requirements that implement the controls:

| Hazardous situation | Risk-control requirements (RC) |
|---|---|
| Wrong position read off an image | SR-VIEW-05, SR-VIEW-07 |
| Old image taken as current | SR-RUN-10, SR-RUN-12, SR-RUN-26 |
| ROI sum from the wrong image or frame | SR-MEAS-04 |
| False precision | SR-MEAS-01, SR-MEAS-03 |
| App hangs or crashes on a long or failing run | SR-RUN-09, SR-RUN-10, SR-RUN-03 |
| Status missed by colour-blind or screen-reader users | SR-A11Y-04 |

## 7. Not required (explicit exclusions)

Stated so that their absence is not read as a gap in verification:

- Loading scenes, and saving anything other than the report export (SR-EXPORT-01 … 05): no image files, no measurement files, no remembered folder or setting (the Windows save dialog keeps its own recent-file and last-folder history, as for every application that uses it).
- Creating measurements from the keyboard — known gap AN-01, planned.
- High-contrast mode — known gap AN-03, planned.
- Four-channel Anger position readout, spatial SiPM pitch and optical light sharing, including crosstalk-dependent crystal identification, are excluded from the current direct-crystal readout. Correlated cascades are modelled for Co-60 and Na-22 only; Ir-192's cascades are emitted as independent lines.
- Any physics accuracy claim — the engine's own suite covers it (VV.Studio §1 (GCAM-SW-VVR, 1. Scope and intended use)).

## 8. Re-evaluation (§5.2.5–5.2.6)

- This baseline was written after the code; each row was checked against the code on 2026-10-01 and has at least one verification entry in the matrix. Rows whose evidence is *partial* are listed in VV.Studio §4 (GCAM-SW-VVR, Coverage summary).
- A change that adds or alters behaviour adds or edits its row here **and** its matrix row in VV.Studio in the same commit.
- When the risk table in VV.Studio §6 changes, re-check §6 above.

**Physical readout (TODO-40, 2026-10-08).** SR-DET-05 … 09, SR-WAVE-09 … 11 and SR-RUN-30 … 34 govern the experimental four-output readout. Where an earlier row describes the analytic detector response, true-deposit traces, windows from that response or stripping (e.g. SR-RUN-16 / -21, SR-WAVE-01 / -02 / -04 / -06, the Spectrum / Imaging window rows), it applies to DirectCrystal acquisition; a physical acquisition uses stored measured energies selected by explicit inclusive keV bounds, with the legacy response, energy-only pile-up and stripping unavailable. SR-RUN-12's frozen inputs include the resolved readout and its preparation identity; SR-RUN-28's counts follow SR-RUN-33 in physical mode.

**Readout presets (TODO-41, 2026-10-12).** SR-DET-05 and SR-RUN-34 were rewritten and SR-DET-10 / -11 added: the physical readout runs on every preset from a shipped, hash-pinned offer record, one calibration per emitted line. The physical window image may apply an opt-in flood correction from the line calibration's own photopeak-assigned flood counts (weight = mean / bin, no clipping); raw counts remain the reported count; an empty correction bin makes the corrected image unavailable, the raw image remaining available. "Displayed" in SR-DET-10 includes text behind a keyboard-focusable help mark whose tooltip opens on hover and keyboard focus and whose automation HelpText carries the same text; the experimental label, a refusal and its short reason, local state, progress and the key cost numbers (RMS, F) stay inline. Because the benchmark is one mono-line point source at a measured 100 000 input hits/s or isolated pulses, Studio's standard isotopes (several emissions) and any scene before an acquisition read the cost table as reference only; the inherited state is defined but no standard scene reaches it.

**Declutter (TODO-44, 2026-10-12).** SR-RUN-26 was reversed (Reset confirms first), SR-DET-10 / -11 dropped the Imaging cost card and table for one record-built help line in the readout panel, and SR-THEME-01 names an icon button. Explanations moved behind help marks under the §2 convention — among them the geometry performance caveat (SR-OPT-02), the count-uncertainty and strip-role details (SR-IMG-05, -07), the processing costs (SR-IMG-06), the focus-sweep details (SR-FOCUS-04; its experimental warning stays inline), the Waveform model note (SR-WAVE-03 / -04 / -07; state, stimulus, clipping and missing-future labels stay inline), the Detector area and gain caveats and the physical-chain assumptions (§5 Front-end chain); the rows themselves are unchanged. Plan: PLAN.Studio.Declutter.

**Declutter stage 2 (TODO-45, 2026-10-13).** Inputs changed rarely moved into an Options dialog with a draft applied on OK, a locked Close-only view while data exist and a non-default summary in the status bar with a cog badge (SR-RUN-12, -20, -25, -29, SR-CHAIN-01 / -02, SR-DET-01); a refused Start says why in the status bar and leads to the input (SR-RUN-13, SR-OPT-03); the OS title bar was removed for a themed caption (SR-THEME-02, revised, and SR-ENV-01); the left panel became Scene / Optics / Detector tabs (SR-OPT-06, revised) with the optics fields always shown (SR-OPT-02, revised — this reverses the plan's "fields only for Custom"); the Detector face's captions and legend facts moved to the right panel (SR-DET-02 / -03); the readout views became view tabs with an empty state under DirectCrystal (SR-DET-06, SR-WAVE-09); readout preparation reports in the status bar (SR-DET-07); the energy window gained presets with only the measured width labelled benchmark (SR-RUN-33); the measurement tools became icon toggles with a per-row delete (SR-MEAS-05 / -06). SR-A11Y-03's tool access keys were not carried over to the icon toggles (VV.Studio AN-16). Plan: PLAN.Studio.Declutter.

**Physical readout with the ambient field (TODO-42 stage A, 2026-10-13).** SR-RUN-34 now admits source scenes with the ambient field on both bounds and refuses field-only scenes (no field calibration line is offered) and, until stage B, nonzero BSR (SR-RUN-22); SR-RUN-13 notes the refusal; SR-RUN-29 and the ambient-bound input row describe the physical path's module construction (the all-face bound becomes an unmasked all-face module approximation); SR-DET-05 states that only source lines calibrate and that off-energy positioning of field deposits is a declared, unvalidated approximation. Evidence: readout-field-v1 (16 outer seeds, exact count identities and eight derived-tolerance checks pass). Stage B (2026-10-13) admits BSR with the physical readout (SR-RUN-22, -34; the input row): fresh module-transported histories keep their own pixel and sites, the pre-trigger ratio and schedule are unchanged; evidence readout-field-v2 (16 outer seeds, BSR 0.5 / 1 with and without ambient; exact identities and sixteen derived-tolerance checks pass).

**Access keys and the Options Enter key (2026-10-13).** The measurement tools' access keys Alt+P / D / A / R were restored as window key bindings on the icon toggles, and SR-A11Y-03 now states the shipped Enter behaviour (Start is not the default button) instead of "Enter runs" (VV.Studio AN-16, resolved). The Options dialog lost its default button (SR-RUN-12): Cancel had been the default, so Enter in a field discarded the draft; the author: Enter in a field must neither apply nor discard a draft.

## Waveform and shared detection chain (2026-10-02)

These requirements replace the earlier exclusion of the Waveform workspace and read-only detection-chain selection. Four-channel position reconstruction remains excluded; the scope models the summed energy channel. Implementation verification is recorded separately in VV.Studio.Waveform (GCAM-SW-VVR).

| ID | Requirement | Verification |
|---|---|---|
| SR-CHAIN-01 | A shared scintillator / photosensor / preamp selection, made on the Options dialog's Detector tab, shall be frozen at Start in the acquisition's detector settings. Measurement, spectrum bands/grouping/label, isotope windows/calibration and waveform shall use that acquired chain. | WaveformServiceTests.Response_AgreesAcrossWaveformSpectrumAndImaging; MixedFieldCalibration_UsesAcquiredChainAndInvalidatesRatios (Evidence); OptionsViewModelTests.Draft_ChangesOwnerOnlyOnApply |
| SR-CHAIN-02 | Chain editing shall be disabled during acquisition, during a readout preparation, while acquired data exist and under the experimental physical readout (which does not use the chain; a non-default chain is then marked inactive in the Options summary); retained data shall never be remeasured with other settings. | WaveformWorkspaceTests.Chain_LockedWithData_ScopeUsesAcquiredChain; ActiveAcquisition_DisablesPhysicalChainChanges; OptionsViewModelTests.PhysicalMode_DisablesOnlyLegacyFields_AndMarksLegacyDifferencesInactive, LockedOptions_AreCloseOnly_AndDoNotApply; OptionsRenderTests.OptionsTabs_MovedIdsAndLocks_UseTheDraft_AndCloseOnlyWhenLocked (chain selectors in the dialog) |
| SR-CHAIN-03 | Studio shall explicitly map GAGG(Ce), NaI(Tl), LYSO, CsI(Tl) and BGO to their transport material. GAGG shall remain the default; other scintillators are what-if comparisons. CsI(Tl) shall use validated CsI host transport data with the Tl activator omitted. Unmapped scintillators shall fail before acquisition. | CsIMaterialTests; WaveformServiceTests.Chain_SetsTransportMaterialWithoutChangingEngineDefaults; Chain_RejectsUnsupportedMaterial; AcquisitionServiceTests.CsISelection_IsCapturedByTheRealAcquisition; WaveformWorkspaceTests.CsI_IsOfferedAndRequiresResetForANewAcquisition |
| SR-WAVE-01 | Waveform shall display ADC and integer-shaper traces (CR-RC Q12 output displayed as fractional codes) of the array-wide energy channel, sharing a navigable time axis, in a default 10 µs window with 20% pretrigger. Event labels appear above the ADC plot only; both plots retain their event marker lines, with full details in the event list. | WaveformServiceTests.Window_PreservesRequestedLengthAndPrecedingPulseAcrossPixels; offscreen real10us renders, shared navigation and upper-only label assertions |
| SR-WAVE-02 | A user shall select latest / next / zero-based acquired event index. Window events shall retain index, pixel, true deposit and absolute acquired time. Default scope time shall preserve recorded arrivals. | WaveformWorkspaceTests.HiddenScope_DoesNotGenerateAndHeldTriggerReusesWindow; WaveformServiceTests.RetainedMonteCarloEvents_PreserveAssociationAndTime |
| SR-WAVE-03 | Rate study shall deterministically re-space retained events in order at a labelled simulated detected rate for the energy channel; it shall not alter events, counts, live time, other workspaces or the acquisition state. | WaveformServiceTests.RateStudy_IsDeterministicAndLeavesAcquisitionUntouched; WaveformWorkspaceTests.ScopeFailure_IsVisibleAndLocalControlsKeepTheAcquisition |
| SR-WAVE-04 | Realistic stimulus amplitudes shall apply acquired pixel gain and the shared index-addressed chain response once, with no second intrinsic raster smearing. Labels shall distinguish the ADC simulation's shaped heights from the analytic MCA. | WaveformServiceTests.Response_AgreesAcrossWaveformSpectrumAndImaging; Rasterizer_MatchesLegacyWithoutSecondIntrinsicSmear |

| ID | Requirement | Verification |
|---|---|---|
| SR-WAVE-05 | Ideal shaper stimulus shall remove chain smearing, analog and ADC noise, and finite rise, retaining fixed pixel gain, selected tail and selected filter (Q12 fractional-state CR-RC or integer trapezoid). | WaveformServiceTests.Ideal_HasNoNoiseOrSmearAndKeepsTail; ideal offscreen renders |
| SR-WAVE-06 | Readouts shall identify acquired chain, photoelectron budget, single-channel FWHM excluding pixel gain spread, pulse time constants, actual integer filter coefficients and effective resolving interval. Trapezoidal energy shall use matched local flat-top calibration and be unavailable for overlap/saturation/partial windows. CR-RC shall report order, Q12 state/Q16 coefficients and explicit per-preset T_sum shaping time (simulation convention, separate from the DCR noise window; not peaking time or hardware evidence), and omit recovered energy until trigger/phase/overlap estimation is validated. | WaveformServiceTests.AcquiredChain_IsUsedAfterPendingSelectionChanges; Readout_SuppressesPartialOverlapAndSaturation; Crrc_PlotsFractionalCodesAndLabelsSimulationTime; CrrcContractTests; RTL contract matrix |
| SR-WAVE-07 | Scope work shall use checked origin-relative sample conversion, explicit requested length, warm-up covering preceding pulse support and filter history, and a total raster allocation cap of 10 million samples. Clipping and missing acquired future shall be labelled; an empty acquired window shall show simulated baseline noise. | ScopeWindow tests; WaveformServiceTests.TimeBase_IsOriginRelativeAtLateAcquisitionTime; Ideal_HasNoNoiseOrSmearAndKeepsTail; MaximumWindow_ReportsWorkerCostAndBoundedPlotSamples (Evidence) |
| SR-WAVE-08 | Scope processing and pyramid preparation shall run on a worker only while the workspace is active. A held covered selection shall reuse its output. Obsolete requests shall be cancelled/discarded and errors shown. Viewport navigation shall not resimulate the trace. | WaveformWorkspaceTests.HiddenScope_DoesNotGenerateAndHeldTriggerReusesWindow; LatestSelectionWinsAndNewAcquisitionCancelsOldScope; ScopeFailure_IsVisibleAndLocalControlsKeepTheAcquisition; shared-axis render assertions |
| SR-WAVE-09 | Physical Waveform shall select a stable measured-event ID and show realised A(x−,y−), B(x+,y−), C(x−,y+), D(x+,y+) analogue pulses and their sum on one acquired time range, with stored trigger/hold instants and signed held ADC codes. The pulse sum shall use retained realised hits and the acquired engine kernel without redrawing fluctuations. The Waveform centre offers Energy sum and Four outputs (A–D) as view tabs in every readout mode, its heading following the view, and a readout-mode change selects that mode's view (Energy sum, or Four outputs under the physical readout): under DirectCrystal, Four outputs shows "Available with the experimental physical readout" and a "Choose readout…" button that selects the left Detector tab and focuses the readout selector; under the physical readout, Energy sum states that the energy sum is the Σ lane of Four outputs. | |
| SR-WAVE-10 | The selected physical conversion shall show raw X/Y, LUT crystal or unknown, calibrated held energy when available, calibration identity and contributor/dominant-hit diagnostics from the stored record used by Spectrum and Imaging. The sum threshold shall state code-equivalent units and the 50 keV-equivalent preset. Continuous digitizer noise and integer-shaper equivalence shall not be implied. | |

| ID | Requirement | Verification |
|---|---|---|
| SR-WAVE-11 | Physical scope generation shall run only while active, reuse covered held-selection output, bound the combined lane/threshold allocation, cancel or discard obsolete requests and preserve shared X navigation. Ideal/rate replay shall be unavailable for physical acquisition. | |

Finite warm-up resets the integer filter locally and is not a claim of acquisition-wide baseline-state equivalence. Ten million samples is an allocation ceiling, not a guaranteed 4 Hz trace-generation rate. The legacy CR-RC F=0 fixture remains bit-exact. The Q12 CR-RC contract is bounded to signed-16 input, A in [0,65536], K in [1,65536], order 1–16 and F 0–12; RTL uses signed 48-bit states/products. This arithmetic bound is not a hardware timing or trigger-efficiency claim.

## Detector and retained-flood focus analysis

| ID | Requirement |
|---|---|
| SR-DET-01 | The reflector-gap editor on the Options dialog's Detector tab uses µm and supplies mm to acquisition. Finite nonnegative gap must be smaller than pending pixel pitch: OK refuses a malformed or too wide draft with the field message; a later pitch change revalidates the gap, and a gap that no longer fits refuses Start with "Options: <reason>", selects the Optics tab and marks the cog (SR-RUN-13). Editing is possible only while no acquired data exist (SR-RUN-12). |
| SR-DET-02 | Detector is the fourth registered workspace. Without acquired data its static gain pattern and face use the pending detector / optics; with data they use the acquired (locked) inputs. The right panel's Detector geometry section names which: "Settings for the next acquisition" or "Acquired settings · locked until Reset" — neutral text, not a warning. |
| SR-DET-03 | The face uses the engine's seeded gain pattern and exact active rectangles with half-gap perimeter and full-gap interior dead regions. It reports geometric active-area fraction ((pitch−gap)/pitch)² as area, independently of acquired counts/live-time rate, material and gain σ/seed. The centre shows the face and its colour bar only; the right panel holds the facts as headed sections of label / value rows — Detector geometry (with the active area), Gain pattern (gain σ / seed, relative-gain range, face size) and Acquired (counts, live-time rate) — top-aligned to its content with standard padding, scrolling when height is constrained. |
| SR-DET-04 | Offscreen fixtures cover Detector before acquisition and with acquired (locked) settings, both themes and 1280×800/1440×900, without creating an HWND or sending desktop input. |
| SR-DET-05 | Detector shall retain DirectCrystal as default and support experimental GAGG FourOutputAnger on any valid geometry. Each calibration line (122.06 / 661.7 / 1332.5 keV) of the pending module shall be planned from the shipped, hash-pinned offer record and the module fingerprint: offered (experimental, at the pinned budget) when the confirmed cell's calibration succeeded on every seed, refused with its literal published reason and population otherwise, locally calibrated (experimental, unvalidated) for an unmatched module, and unavailable when the record is missing or malformed. Preparation shall capture the resolved module geometry, line, policy, budget, seeds and calibration identity; the 12×12 Baseline at 661.7 keV keeps its TODO-40 preparation byte for byte. Local readiness shall follow live cache entries; an evicted line shall read Not prepared and Prepare shall rebuild it. The shell shall not retain evicted preparation buffers. Start shall capture an immutable bank of the required line preparations, used by Detector, Imaging, Spectrum, Waveform and circuit views. With the ambient field or BSR on, only the scene's source lines form the bank; source, BSR and field deposits share one measured pulse stream and are windowed and assigned by the measured record, never by origin or true energy. Positioning field deposits at other energies with a source-line calibration is a declared approximation whose accuracy is unvalidated. Physical component rates are published from the physical producer. IndependentSipm remains outside this Studio implementation. |

| ID | Requirement |
|---|---|
| SR-DET-06 | Detector shall show active SiPM footprints and pitch-cell borders and an illustrative charge-division graph from the same resolved circuit used by the engine. It shall label virtual matched sensors, DPC ratio 0.01, virtual-ground outputs and omitted RC/skew/jitter. Legacy gain and chain controls shall be inactive and disabled in physical mode. The Detector centre offers Face, SiPM, Anger, Calibration and Diagnostics as view tabs in every readout mode, its heading following the view and the mode, and a readout-mode change selects that mode's first view (Face, or SiPM under the physical readout); under DirectCrystal the readout views show "Available with the experimental physical readout" and a "Choose readout…" button that selects the left Detector tab and focuses the readout selector. Under the physical readout the right panel shows Readout network, Measured conversions, Calibration (this session) and Confirmed offers as sections of rows. |
| SR-DET-07 | Explicit Prepare — the accent-styled action of the left Detector tab's Preparation section — shall build an independent calibration on a worker, expose state/progress/cancellation and publish density, ordered markers and LUT boundaries with their identity. While it runs the status bar carries "Preparing <line, stage and history progress>" and the one progress bar; the outcome (ready lines, cancelled, or the failure in the warning colour) stays in the status bar until the next preparation, Start or Reset, while the Preparation section keeps Prepare / cancel, the ready table (keV, time, windowed, scored) and a short state line. Failure shall display the engine reason, also inline, and prevent Start; no direct-assignment fallback is permitted. A bounded cache shall reuse only identical resolved inputs and versioned seed/budget policy. |
| SR-DET-08 | Smoothed calibration density, live raw X/Y counts and LUT-assigned crystal counts shall have distinct labels and denominators. Raw coordinates shall be dimensionless. Independent per-crystal diagnostics shall state their energy, all-energy/window selection, single-crystal truth definition, validation seeds and scored/assigned denominators; training density is not held-out acceptance evidence. |
| SR-DET-09 | Detached readout render fixtures shall cover pending, preparing, ready, failed and acquired displays, sensor/circuit views, flood/LUT overlays and four lanes plus sum in both themes at 1280×800 and 1440×900, without an HWND or desktop input. Desktop verification remains separately gated. |
| SR-DET-10 | Detector and Imaging shall display, per line, the confirmed offer or refusal (population, budget, source record and phase, literal failures, superseded outcome) separately from the local preparation state and its stage / history progress. The Detector Calibration tab shall show a labelled 32-seed mean sensitivity map beside the recorded weakest / median scalar (truth-referenced validation ratios, never flood weights). The readout panel shall carry one line "experimental readout · localisation cost (EV-36)" whose help text is built from the shipped record: the declared benchmark scope, stated as a benchmark reference and not a prediction for the scene; the F range of the module's offered cells; for every F below 1 the Direct path's own bias share with the statement that it does not establish a general Anger advantage; and the sentence "F is an equal-fluence MSE ratio, not a time factor." Nothing is interpolated or mapped to the scene. Imaging's channel section shall show, in physical mode, the window's channel-unavailable refusal, the acquisition's own raw window counts and the opt-in flood-correction control. *Revised 2026-10-12 (TODO-44): the Imaging reference card, full cost table and applicability status were removed — their "inherited" state is unreachable for every standard scene (several emission lines per isotope); the record and its cost rows remain as evidence (EV-36).* |
| SR-DET-11 | Any valid Custom geometry may Prepare. A module matching a confirmed geometry (crystal count, pitch, gap, entrance absorber, backing) shall inherit that geometry's per-line offer or refusal and pinned budget irrespective of the preset label; its localisation help describes that geometry's offered record cells as a benchmark reference, and no imaging cost is claimed for the scene. An unmatched module calibrates locally at 2000 histories per crystal with LatticeGrowth and the adjoint solve, labelled experimental and unvalidated; its localisation help states that no offered cost cells exist; its failure shows its own reason. Held ADC facts shall be acquired once and projected through each captured line calibration, never dispatched by generating line or true energy. |

| ID | Requirement |
|---|---|
| SR-FOCUS-01 | A cancellable worker sweeps the selected channel's retained flood through ImagingProjection.AtFocus with non-cyclic decoding and Studio allocation validation, without transport or calibration and without joining the 4 Hz snapshot publisher. |
| SR-FOCUS-02 | Sweep planes are uniform in 1/z from acquired D+30 to 3000 mm. K is user-selected from 1–4, default 1, independent of scene truth. Across planes, candidates are assigned one-to-one by minimum squared angular displacement (x/z,y/z). |
| SR-FOCUS-03 | A plane-versus-prominence PlotView shows each complete track's curve, sharpest plane and contiguous interpolated raw half-maximum interval. Edge-reaching intervals have censored endpoints and no finite width. Boundary maxima, disconnected half-max modes and unresolved results are labelled. |
| SR-FOCUS-04 | Reset, a new acquisition or edits to channel/window/strip/K cancel and invalidate analysis (optics cannot change while data exist); obsolete responses cannot publish. Results identify the frozen acquisition prefix. Visible text describes lateral-dependent near-field bias, default-optics far-edge censoring from about 700 mm, and the difference between half-max width and uncertainty. Stripped fractional clipped floods are not independent Poisson observations. |
| SR-FOCUS-05 | Optional external surface range is separate from decoder focus and appears as a focus-curve marker. Only the explicit Use as focus action changes decoder focus. No fusion verdict, significance or automatic focus change is supplied. |

The focus tool is descriptive engineering analysis, with no calibrated depth accuracy or confidence interval. Conditions and verification limits are in VV.Studio.Detector (GCAM-SW-VVR).

## Report export (TODO-46, 2026-10-15)

| ID | Requirement |
|---|---|
| SR-EXPORT-01 | An "Export report…" document icon button (Alt+E) sits in the top bar beside Reset. It is enabled only while a snapshot exists. It opens an in-window "Export report" popup listing HTML (the default), PDF, Word (.docx) and Event data (CSV), each with a one-line description and each available while a snapshot exists; Cancel and Escape close it. Export… opens the standard save dialog titled "Export report", in Documents, with the name `gcam-report-<yyyyMMdd-HHmmss>.<html / pdf / docx>` or `gcam-events-<yyyyMMdd-HHmmss>.csv`, the format's filter and the dialog's own overwrite prompt. When a declared sidecar beside the chosen path already exists, Studio asks once before replacing it. Cancel writes nothing. No folder or other setting is remembered. *Revised 2026-10-15 (stage 2): document icon, popup and formats.* |
| SR-EXPORT-02 | The report is captured when the command runs, before the popup opens, from the published snapshot and the view models; the files are built from that capture after the path is chosen, fully in memory. HTML, PDF and Word carry the same content in the same order: the PDF draws the text, tables and figures as vectors with the acquired configuration as a monospace appendix (PDFsharp); the Word file uses heading styles, bordered tables and PNG figures, with the configuration as an appendix. Nothing is recomputed. The HTML report is one self-contained file with inline CSS and inline SVG figures, and no script or external reference. It holds: a header with the notice "Simulation — not a measurement", the export time and the Studio version; the scene, with the acquired configuration in which the ambient spectrum's tables are replaced by their entry counts (the full tables are in the scene file); the acquisition; the Imaging heat map in Studio's colormap with the estimate and found peaks; the spectrum with its windows and line markers; and the Imaging measurements. The same inputs give identical bytes. |
| SR-EXPORT-03 | Beside every export, Studio writes `<name>.scene.json`: the configuration the acquisition ran with, including the resolved ambient spectrum, serialised by the engine's configuration loader. It records the inputs and does not promise to reproduce the run. When the readout is not the direct crystal assignment, the report states that the captured calibration bank is not in the file. |

| ID | Requirement |
|---|---|
| SR-EXPORT-04 | Studio writes only the confirmed path and its declared sidecars. All destinations are opened before any is truncated. A failed write restores the earlier bytes, or removes a file the export created, and shows "Report export failed" with the reason. The acquisition is not changed. |
| SR-EXPORT-05 | Event data writes `<name>.csv`: one row per retained event, a header row, invariant culture, CRLF, no byte-order mark. Under the direct crystal assignment the columns are index, arrival_time_s, pixel_x, pixel_y, deposit_keV and measured_keV (the value the spectrum histogrammed for that event, replayed from the spectrum's index-keyed measurement with no new randomness; empty for a pulse absorbed in a pile-up group). Under the physical readout they are index, hold_time_s, trigger_time_s, A, B, C, D (integer 14-bit codes), raw_x, raw_y, crystal (−1 when unassigned) and energy_keV (empty when unassigned). Beside it, `<name>.meta.json` names the readout and what it means (the direct assignment has no light, network or ADC stage), each column's meaning and unit, the sibling files, the scene and acquisition summaries, and the configuration with the ambient tables as entry counts. *Added 2026-10-15 (stage 2, X8).* |

## Readout comparison (TODO-47, 2026-10-15)

| ID | Requirement |
|---|---|
| SR-CMP-01 | A fifth workspace, Compare (Ctrl+5), shows one physical-readout acquisition read two ways from the same retained inputs: "Direct crystal · ideal deposit energy" (the largest summed-deposit pixel and the unsmeared total deposit of each consumed input) and the four-output Anger readout as acquired. It starts no transport, calibration or second acquisition and runs no legacy measurement chain on the direct path. |
| SR-CMP-02 | Each column shows its flood map above its reconstruction, with truth ○ and found ◇ markers as in Imaging; both paths use Imaging's current decoder, focus, channel and explicit keV window, and a window uses only its captured calibration line (an unavailable line is stated, never replaced by another). A toggle replaces the flood row with the assignment-disagreement map — per direct pixel, different / eligible conversions, an empty denominator shown as unavailable — while the reconstructions stay. A spectrum strip overlays both paths' raw counts on identical bins with the shared window and an overflow count. |
| SR-CMP-03 | The side panel states explicit populations: histories, held, assigned, unknown, multi-contributor, no-dominant, distinct-dominant and never-dominant inputs, finalized and unresolved no-contribution counts; the same-crystal share is Σ same / Σ eligible over single-significant-contributor conversions; each path's estimate, its truth offset (mm, °) and the path separation. |
| SR-CMP-04 | The reference cost table shows the 1 m offer-cost record rows for the captured preparation, always marked as a benchmark reference, with the listed differences between this scene and the benchmark (isotope / line, distance, position, rate, module). |
| SR-CMP-05 | States are explicit: under the direct assignment, "Available with the experimental physical readout" and "Choose readout…" (no comparison numbers); before an acquisition, "Start an acquisition to compare"; without held, assigned or window events, the direct image stays and the reason is shown; while updating, the last consistent pair stays with its prefix time. The comparison runs on its own coalescing worker only while Compare is active and never delays the acquisition or the Imaging / Spectrum refresh. The report export states that Compare is not included. |

Limits stated with Compare: never-dominant inputs are not a conversion-loss estimate; no-contribution counts finalize only after pulse support expires; single-significant-contributor agreement is not an isolation proof; the ideal direct energy has no detector FWHM and the Anger FWHM is not estimated here; cost rows are benchmark references, not time factors or predictions for a Studio scene.